Hypertension

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Clinical Modifiers of Salt Sensitivity: A Metaregression and Risk Translation Study

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BACKGROUND: Hypertension guidelines generally recommend uniform sodium restriction although determinants of salt sensitivity, or sodium-blood pressure responsiveness, remain incompletely quantified. We estimated the population sodium-blood pressure dose-response, identified clinical modifiers, quantified high salt sensitivity prevalence, and derived the number needed to restrict sodium intake to prevent 1 cardiovascular event across clinical strata.

METHODS: PubMed, Embase, and Cochrane were searched from inception to April 2025 for sodium intervention trials and observational studies. Blood pressure response was standardized per 50-mmol/d sodium difference. Random-effects metaregression evaluated clinical modifiers. A highly salt-sensitive phenotype was defined a priori as a systolic blood pressure (SBP) decrease ≥3 mm Hg per 50-mmol/d reduction.

RESULTS: We included 160 studies (255 estimates; n=16 443). Each 50-mmol/d higher urinary sodium excretion was associated with 1.76-mm Hg higher SBP (95% CI, 1.54–1.98). Age and baseline SBP were the strongest modifiers (0.41 mm Hg per 5 years; 0.23 mm Hg per 5 mm Hg). Salt-sensitive phenotype prevalence rose from 12.5% (age <40 years) to 56.5% (>60 years) and from 16.3% (SBP <120 mm Hg) to 54.1% (≥130 mm Hg). Under a policy-relevant 100-mmol/d sodium reduction, number needed to restrict ranged from ≈1334 in younger, normotensive, lean individuals (age <60 years, SBP <130 mm Hg, and body mass index <25 kg/m^2) to ≈87 in older adults with hypertension and elevated body mass index (age ≥60 years, SBP ≥130 mm Hg, and body mass index ≥25 kg/m^2), a 15-fold gradient.

CONCLUSIONS: Salt sensitivity varied across age, SBP, and body mass index strata. These findings suggest that routinely measured clinical characteristics may help identify populations with greater expected blood pressure response to sodium reduction. ***(Hypertension.*** **2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.126.27410.)** • **Supplement Material**.

Key Words: antihypertensive agents ■ blood pressure ■ cardiovascular diseases ■ hypertension ■ sodium, dietary

Hypertension is a leading modifiable risk factor for cardiovascular morbidity and mortality worldwide.[1,2] Among lifestyle-based strategies, dietary sodium reduction is a cornerstone of blood pressure (BP) management and is consistently recommended in clinical guidelines.[3,4] However, BP responses to changes in sodium intake vary widely: some individuals experience marked BP lowering with sodium reduction, whereas others derive little benefit despite comparable sodium change; conversely, increasing sodium intake can elicit substantial BP rises in susceptible individuals.[5] Salt sensitivity, defined as the change in BP per unit change in dietary sodium intake, provides a physiological framework to characterize this interindividual variability.[6]

Both the 2024 European Society of Cardiology guidelines[3] and the 2025 American Heart Association/American College of Cardiology guideline for high BP[4] recommend dietary sodium restriction below 87 to 100 mmol per day (equivalent to 2000–2300 mg/d; 1-mmol Na=23.0 mg) as a universal lifestyle intervention for patients with hypertension, irrespective of age, baseline BP level, or metabolic status. In contrast

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NOVELTY AND RELEVANCE

What Is New?

Blood pressure responses to sodium are not uniform but vary systematically by age, baseline blood pressure, and body mass index.

We translated this heterogeneity into clinically interpretable differences in projected cardiovascular benefit.

What Is Relevant?

Sodium reduction may have the greatest blood pressure benefit in older adults and people with hypertension or elevated body mass index.

These routinely measured characteristics could help prioritize intensive sodium-reduction counseling.

Clinical/Pathophysiological Implications?

Salt sensitivity may represent an integrated marker of vascular aging, impaired renal sodium handling, and metabolic burden, supporting future precision-oriented sodium guidance.

Nonstandard Abbreviations and Acronyms

BMI	body mass index
BP	blood pressure
DBP	diastolic blood pressure
IPD	individual participant data
MAP	mean arterial pressure
NNR	number needed to restrict
SBP	systolic blood pressure

to antihypertensive pharmacotherapy, which increasingly adopts a precision-medicine orientation with treatment selection guided by phenotype, comorbidities, and response biomarkers,[7] dietary sodium recommendations have remained uniform mandates, unadjusted for individual susceptibility. This constitutes a critical evidence gap: if salt sensitivity follows a predictable gradient across readily measurable clinical characteristics, individualized sodium targets could substantially increase the efficiency and efficacy of lifestyle-based BP management.

Beyond its relevance to BP control, salt sensitivity has been associated with increased cardiovascular risk. Longitudinal studies have demonstrated that individuals with greater salt sensitivity have higher risks of cardiovascular events, mortality, and incident hypertension, even among those who are normotensive at baseline.[8–10] These findings suggest that salt sensitivity reflects an underlying susceptibility that precedes the development of overt hypertension. Despite its established pathophysiological basis, implicating impaired renal sodium handling and vascular dysfunction, salt sensitivity is rarely assessed in clinical practice, largely because standard measurement protocols require intensive and time-consuming sodium manipulation.[11]

Although sodium intervention trials have examined salt sensitivity across diverse populations, the literature is limited by small sample sizes and reliance on binary thresholds (salt-sensitive versus resistant) that obscure graded physiological variation. Current guidelines recommend population-wide sodium restriction without stratification by susceptibility, and whether readily available clinical characteristics can inform individualized recommendations remains unclear. Sodium restriction, thus, represents a domain of cardiovascular medicine where precision-medicine principles have not been systematically applied. We, therefore, conducted a preregistered systematic review and metaregression to (1) quantify the standardized sodium-BP dose-response; (2) examine study-level modifiers of sodium-BP responsiveness, including age, baseline BP, and body mass index (BMI); (3) describe the projected prevalence of higher salt sensitivity across clinically defined population strata; and (4) translate these stratum-specific sodium responsiveness estimates into clinically actionable number needed to restrict (NNR) projections across age-systolic BP (SBP)-BMI strata, providing a quantitative foundation for precision-oriented sodium policy recommendations.

METHODS

This study is a meta-analysis of aggregate data drawn entirely from previously published studies; no individual participant data (IPD) were collected. Data generated by the present study (ie, the extracted study-level data set and analytic code) will be made available to other researchers for purposes of reproducing the results or replicating the procedure, upon reasonable request to the corresponding author. The preregistered protocol (International Prospective Register of Systematic Reviews: CRD420251035133)[12] and analytic code are available from the corresponding author on reasonable request. We conducted a systematic review and meta-analysis in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.[13]

Data Sources and Searches

We searched PubMed, Embase, and Cochrane Central Register of Controlled Trials from database inception to April 2025 without language restrictions. Search terms combined concepts related to sodium intake (eg, sodium and salt) and BP responses (eg, salt sensitivity and BP). Detailed search strategies for each database appear in Appendix S1.

Study Selection

Studies were eligible if they (1) enrolled human participants aged ≥18 years; (2) reported an intervention or controlled exposure that altered dietary sodium intake (eg, sodium loading, restriction, or crossover protocols); (3) measured SBP or diastolic BP (DBP) or mean arterial pressure (MAP) under both high- and low-sodium conditions; (4) reported or allowed derivation of the change in sodium (ΔNa) and corresponding change in BP (ΔBP); and (5) provided sufficient information to derive a measure of variability for ΔBP (eg, SD, SE, or CI). Studies were not restricted by hypertension status or subtype; those enrolling normotensive participants, patients with hypertension (including those with resistant hypertension), or mixed populations were all eligible.

We excluded studies that (1) lacked pre-post–BP measurements; (2) evaluated only acute hemodynamic responses under anesthesia or invasive procedures, which are physiologically distinct from chronic salt sensitivity; or (3) were reviews, editorials, conference abstracts without extractable data, non-English articles, or duplicate reports from the same cohort, in which case the most complete data set was retained.

Eligible designs included randomized crossover trials, parallel-group randomized controlled trials, nonrandomized controlled studies, and cohort studies with paired high- and low-sodium assessments. Cross-sectional design not permitting derivation of within-group sodium-BP change was excluded.

Data Extraction and Quality Assessment

Three reviewers (S.-H.C., C.-Y.Y., and Y.-P.C.-L.) independently extracted study characteristics, participant demographics, sodium intervention details, study design, and outcome measures using a standardized form. Discrepancies were resolved through discussion or consultation with a fourth reviewer (H.-M.C.). Study selection is summarized in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram (Figure 1). In studies utilizing 24-hour urine collection, completeness was acknowledged as reported by each primary study; when urine adequacy criteria were not explicitly defined, this was indicated in the risk-of-bias evaluation within the measurement-of-exposure domain.

The risk of bias of each included study was assessed according to study design, using the RoB 2 tool and the Newcastle-Ottawa Quality Assessment Scale for interventional and observational cohort studies. Potential publication bias and small-study effects were evaluated using funnel plot asymmetry and the Egger regression test.[14,15]

The certainty of evidence was assessed using the GRADE framework (Appendix S4).

Definition of Salt Sensitivity

For each study or study arm, salt sensitivity was defined as the ΔBP per 50-mmol/d higher sodium intake.[6] The primary outcome was ΔSBP, with ΔDBP and ΔMAP evaluated as secondary outcomes. Changes in BP were calculated as the difference between high- and low-sodium conditions.

The ΔNa was defined as the absolute difference in sodium intake between high- and low-sodium conditions. Sodium intake was quantified using either 24-hour urinary sodium excretion or dietary assessment. To maximize available evidence while prioritizing more objective exposure assessment, we applied a hierarchical sodium ascertainment strategy: 24-hour urinary sodium was used whenever available, and dietary sodium intake was used only when urinary data were not reported. We refer to this harmonized approach as the combined sodium estimation. To validate this approach, we prespecified a sensitivity analysis comparing effect estimates derived from urinary versus dietary assessment methods to ensure consistency in direction and magnitude. Research utilizing 24-hour urine sodium measurements was considered the key stratum; dietary sodium findings are presented as a robustness study to verify directional consistency, as urinary sodium assessment is more dependable for evaluating sodium intake.

To standardize effect sizes to a common scale, salt sensitivity was operationalized as the pooled regression coefficient describing the change in BP per 50-mmol/d higher sodium intake, derived from a weighted random-effects metaregression in which the observed ΔBP served as the response variable (weighted by its own sampling variance), and ΔNa served as a study-level exposure covariate, analogous to the other study-level moderators (eg, age, baseline SBP, and BMI) evaluated in subsequent models. Consistent with standard regression convention, in which predictor variables are treated as measured without error, ΔNa was treated as a fixed covariate; only the sampling variance of ΔBP was propagated into model weights. Regression coefficients were scaled by 50 to express salt sensitivity as the predicted ΔBP per 50-mmol/d sodium difference, with positive values indicating higher BP with higher sodium exposure.

When the SDs of interested outcomes were not reported, they were derived using SDs from the high- and low-sodium conditions. For studies lacking paired SDs, the standard formula for correlated measures was applied: $SD_\Delta = \sqrt{SD_1^2 + SD_2^2 - 2rSD_1SD_2}$, assuming a correlation coefficient of r=0.5.[16] SEs were computed as $SE = SD_\Delta/\sqrt{n}$, and corresponding 95% CIs were derived.

Statistical Analysis

Random-effects meta-analyses and metaregression models were fitted using restricted maximum likelihood, with between-study heterogeneity quantified by the I^2 statistic and τ^2. Because multiple effect estimates could originate from the same study, we accounted for within-study dependence by clustering effect sizes at the study level using robust variance estimation.[17]

Potential effect modifiers were examined using univariable and multivariable random-effects metaregression. Prespecified covariates included age, sex distribution, BMI, baseline SBP/DBP/MAP, prevalence of hypertension or diabetes, kidney function, and ethnicity. Continuous covariates were modeled per prespecified increments (eg, age per 5 years and sex per 10 percentage points). Knapp-Hartung adjustments were applied.[18]

Prespecified subgroup analyses were conducted to describe how salt sensitivity varied across clinically relevant categories.

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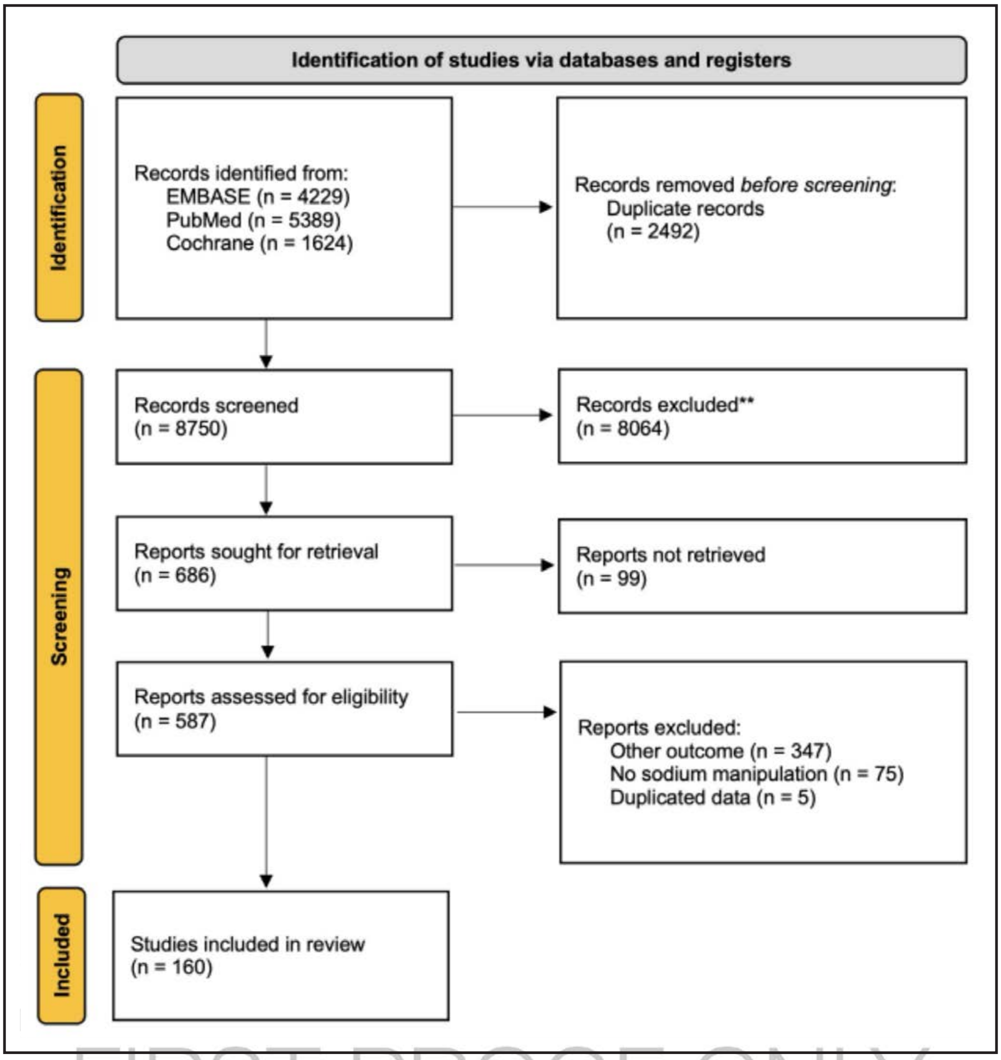

Figure 1. Study selection flow diagram.
The flow diagram illustrates the identification, screening, eligibility assessment, and inclusion process for studies included in the review. Records were identified from Embase, PubMed, and Cochrane databases. After the removal of duplicate records, titles and abstracts were screened, followed by full-text assessment for eligibility. Studies were excluded primarily because they reported outcomes other than blood pressure response, did not include sodium manipulation, or contained duplicated data. A total of 160 studies were included in the final review.

Studies were stratified into predefined groups based on age (<40, 40–60, and ≥60 years), proportion of women (0%–33%, 33%–67%, and ≥67%), BMI (<25, 25–30, and ≥30 kg/m^2), prevalence of hypertension or diabetes (0%–50% versus ≥50%), kidney function (estimated glomerular filtration rate <60 versus ≥60 mL/min per 1.73 m^2), racial composition (0%–50% versus ≥50% Asian; 0%–50% versus ≥50% Black), and intervention type (sodium loading versus sodium reduction). Random-effects forest plots and metaregression visualizations were generated using the same prespecified subgroup definitions.

We fitted prespecified sequential multivariable metaregression models to evaluate whether associations between sodium-BP responsiveness and key study-level characteristics were robust to adjustment. Candidate covariates were selected a priori based on biological plausibility, clinical availability, and data completeness, with age, baseline SBP, BMI, sex distribution, and Asian proportion included as the main study-level characteristics. Sequential models were used to show how estimates changed after adding related covariates, rather than for automated variable selection. To avoid multicollinearity, we inspected pairwise correlations and variance inflation factors, and no covariate demonstrated problematic collinearity (all variance inflation factors <3; Table S3). For the multivariable models, continuous predictors (age, baseline SBP, and BMI) were mean-centered before analysis to facilitate interpretability of interaction terms and the model intercept; the intercept, thus, represents the predicted sodium-SBP slope at the sample mean values of all covariates. We first fit each covariate individually, followed by sequential adjustment models adding 1 variable at a time, prioritizing covariates with strong univariable effects (age, baseline SBP, and BMI). Analyses were repeated after imputing missing covariates via chained equations.

To examine the potential contribution of concurrent potassium changes to the observed sodium-BP associations, we conducted 4 supplementary analyses, restricted to studies that reported change in urinary potassium excretion (ΔK). First, we tested whether ΔK modified the sodium-SBP slope by incorporating a ΔNa×ΔK interaction term in the metaregression framework. Second, we examined confounding by adding ΔK as an independent covariate in sequential adjustment models: (1) ΔNa only, (2) ΔNa+ΔK, (3) ΔNa+age+BMI+baseline SBP, and (4) the full model additionally including ΔK. Third, among effect estimates with both ΔNa and ΔK data, the urinary sodium-to-potassium ratio was examined as an alternative exposure. Fourth, we examined whether baseline urinary potassium excretion modified salt sensitivity, using the same interaction framework as other study-level modifiers.

To enhance clinical interpretability, we conducted a model-based risk-stratified prediction analysis using representative values of age, baseline SBP, and BMI. Predicted ΔSBP was first estimated per 50-mmol/d sodium reduction across prespecified joint strata defined by age (<60 versus ≥60 years), baseline SBP (<130 versus ≥130 mm Hg), and BMI (<25 versus ≥25 kg/m^2), yielding 8 clinically relevant groups. Representative values for each category were selected from the study-level data distribution and fixed for prediction. To further characterize the joint gradient of sodium responsiveness, we additionally constructed an age×SBP prediction matrix (Appendix S4).

For cardiovascular risk translation, ΔSBP estimates derived per 50-mmol/d sodium reduction were linearly rescaled to reflect a policy-relevant scenario: reduction from the estimated global mean sodium intake (≈187 mmol/d),[19] to the WHO recommended target of 87 mmol/d,[20] corresponding to a sodium reduction of ≈100 mmol/d. This scalar approach assumes a linear dose-response within the observed range, consistent with the restricted cubic spline analyses that supported approximate linearity over the range of observed sodium intakes.

Relative cardiovascular risk reduction was derived from a contemporary large-scale meta-analysis,[21] which reported an ≈2% lower cardiovascular disease risk per 1-mm Hg lower SBP. Assuming a log-linear association, predicted relative risk after sodium reduction was calculated as 0.98^ΔSBP. Baseline 10-year cardiovascular disease risk was estimated using SCORE2 based on stratum-representative age and SBP values.[22] The NNR was calculated as the reciprocal of the 10-year absolute risk reduction, where absolute risk reduction=baseline risk×(1−relative risk).

To characterize the population-level distribution of salt sensitivity, we conducted a simulation using all study arms with adequate data for ΔSBP and ΔNa (Appendix S4). In addition, we estimated the projected prevalence of a higher salt-sensitive phenotype across key study-level subgroups, including categories defined by age, female proportion, BMI, and baseline SBP. This phenotype was defined a priori as an SBP increase of ≥3 mm Hg per 50-mmol/d higher sodium intake, a pragmatic threshold informed by prior population-based salt sensitivity literature and broadly concordant with historically reported salt-sensitive and salt-resistant proportions.[9,10,23] These analyses complemented the metaregression by illustrating population-level differences in salt sensitivity.

To assess whether temporal trends in BP measurement methodology affected the primary conclusions, we conducted a sensitivity analysis restricting analyses to studies published in or after 2003 (post-JNC7 era) and, separately, to studies published in or after 2005 (post-JNC7 with a 2-year dissemination lag).

All analyses were performed in R (version 4.5.1) with the metafor package.

RESULTS

Study Selection and Risk of Bias

Figure 1 summarizes the study selection process. Overall, 160 studies contributed 255 effect estimates, representing 16 443 participants, and were included in the quantitative synthesis. Across studies, the mean age ranged from 22 to 73 years, the proportion of women ranged from 0% to 100%, and baseline SBP ranged from 98 to 187 mm Hg.

Risk of bias assessments are summarized in Figures S11 and S12. Overall, risk of bias was generally low to moderate across designs; the certainty of evidence for BP outcomes in randomized trials was not downgraded for risk of bias in the GRADE assessment. Using the GRADE approach, certainty differed by study design: in randomized controlled trials, certainty was high for SBP and DBP, and moderate for MAP. In contrast, evidence from nonrandomized studies was of lower certainty, rated as low for SBP and DBP, and low for MAP.

Overall Effect of Sodium Intake on BP

All effect estimates were standardized to represent the BP difference per 50-mmol/d difference in sodium intake. Using the 24-hour urinary sodium as the primary exposure metric, the metaregression estimate for ΔSBP was 1.76 mm Hg per 50-mmol/d sodium difference (95% CI, 1.54–1.98; Table S1). To assess the robustness of these findings, we conducted univariable metaregression analyses for ΔSBP, ΔDBP, and ΔMAP stratified by sodium assessment method (Figure 2; Table S1). The direction and magnitude of associations were highly consistent across measurement approaches. As a robustness analysis, we stratified estimates by sodium assessment method (Table S1). Studies using a combined estimation approach produced an estimate of SBP response of 1.78 mm Hg (95% CI, 1.57–2.00; I^2=76.2%), while dietary sodium-based estimates yielded a directionally consistent, slightly larger effect of SBP response of 1.92 mm Hg (95% CI, 1.24–2.61; I^2=90.6%) per 50-mmol/d sodium difference. This concordance supports the use of 24-hour urinary sodium as the primary metric and validates the combined estimation approach for analyses where urinary data are unavailable.

Restricted cubic spline analyses suggested that the sodium-SBP association was broadly compatible with a linear approximation over the range where most data were observed, supporting the interpretability of the primary linear model estimates (Figure S7). The nonlinear spline terms were statistically significant (Wald test P<0.001), suggesting a modest deviation from strict

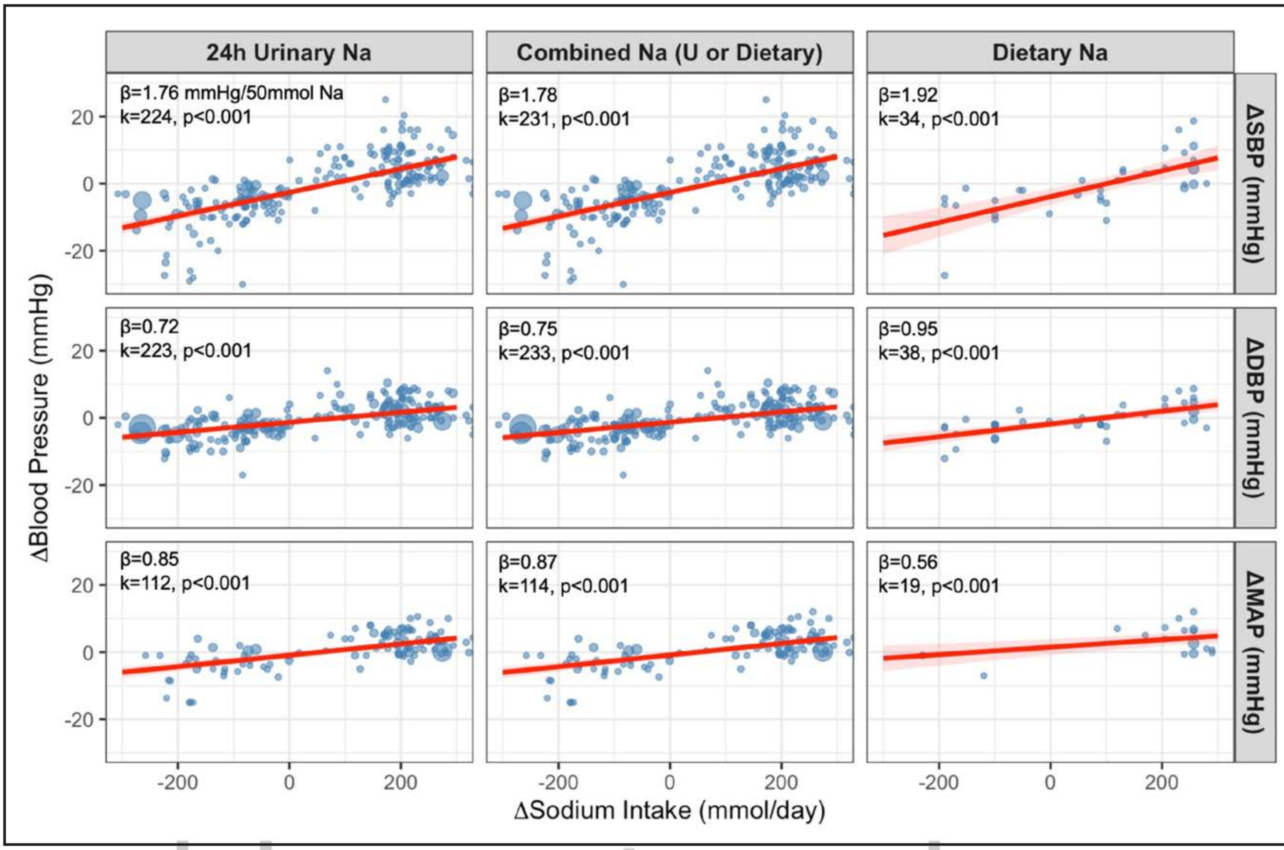

Figure 2. Metaregression of blood pressure changes on sodium change.
Metaregression analyses showing the association between changes in sodium and changes in blood pressure across studies. Columns represent sodium exposure metrics, including 24-hour urinary sodium, combined urinary or dietary sodium, and dietary sodium. Rows represent changes in systolic blood pressure (SBP), diastolic blood pressure (DBP), and mean arterial pressure (MAP). Each point represents an individual study estimate, with point size reflecting study weight. Red lines indicate fitted metaregression slopes with corresponding confidence bands. The β coefficient represents the estimated blood pressure change per 50-mmol/d change in sodium.

linearity at the extremes of sodium reduction and loading. However, within the range where most data were concentrated, the relationship remained approximately monotonic, supporting the interpretability of the primary linear model estimates.

Effect Modifiers on Salt Sensitivity

Metaregression analyses identified several study-level characteristics that significantly modified the SBP response to sodium intake (Table 1; Figure S1). Age was the strongest modifier: each 5-year increase corresponded to a 0.41-mm Hg larger ΔSBP per 50-mmol/d sodium difference (P<0.001). Higher female proportion and higher baseline BMI also amplified salt sensitivity (0.12 mm Hg per 10% women and 0.13 mm Hg per BMI unit, respectively).

Baseline BP demonstrated a similarly strong modifying effect. For every 5-mm Hg higher initial SBP and DBP, ΔSBP increased by 0.23 and 0.30 mm Hg, respectively (both P<0.001). A higher prevalence of hypertension further increased salt sensitivity, whereas diabetes prevalence showed no clear interaction. Lower estimated glomerular filtration rate suggested a borderline trend toward greater salt sensitivity (P=0.074).

These modifying effects were clinically evident when stratified into commonly used risk categories (Figure 3). The estimated SBP response was modest among younger individuals (<40 years: 0.94 mm Hg per 50 mmol/d), those with BMI <25 kg/m^2 (1.11 mm Hg) and baseline SBP <120 mm Hg (0.95 mm Hg) but increased substantially among older adults (≥60 years: 6.03 mm Hg), individuals with overweight or obesity (≈3.2–3.4 mm Hg), and those with baseline SBP ≥130 mm Hg (3.59 mm Hg). This graded pattern underscores the marked heterogeneity in sodium-related BP susceptibility across clinically identifiable subgroups.

Sodium-SBP associations did not differ substantially between crossover and pre-post–designs, nor between randomized and nonrandomized studies. In contrast, sodium-loading protocols yielded a modestly smaller BP response compared with sodium-reduction trials (P=0.037). Other cardiometabolic and study-level factors showed weaker or inconsistent modification (Table 1).

Table 1. Univariable Metaregression for Effect Modification of the SBP Response to Sodium Change

Modifier	k (effects number)	β interaction	95% CI	Knapp-Hartung *P* value	Δτ2	R2 analog
Age (per 5 y)	231	0.410	0.315 to 0.504	<0.001	7.545	27.4
Female (per 10%)	223	0.119	0.032 to 0.207	0.008	0.853	3.1
Initial BMI	166	0.128	0.047 to 0.210	0.002	1.641	7.3
Initial eGFR	48	−0.031	−0.066 to 0.003	0.074	3.352	9.6
DM history, %	27	0.096	−0.088 to 0.280	0.293	−0.196	−0.7
Hypertension history, %	182	0.147	0.098 to 0.197	<0.001	6.025	18.9
Hypertension family history, %	23	0.163	−0.051 to 0.378	0.127	2.106	6.1
Initial SBP (per 5 mm Hg)	212	0.227	0.164 to 0.290	<0.001	6.878	25.1
Initial DBP (per 5 mm Hg)	210	0.300	0.209 to 0.391	<0.001	5.517	20.1
Initial MAP (per 5 mm Hg)	81	0.287	0.144 to 0.431	<0.001	2.955	15.6
Crossover RCT (vs pre/post)	231	0.392	−0.091 to 0.874	0.111	0.111	0.4
Asian (per 10%)	145	0.054	−0.007 to 0.115	0.084	0.525	1.8
Black (per 10%)	145	0.042	−0.052 to 0.137	0.375	−0.046	−0.2
Salt-loading vs reduction	231	−1.019	−1.977 to −0.061	0.037	2.886	10.5
RCT	231	0.375	−0.087 to 0.837	0.111	0.329	1.2

BMI indicates body mass index; DBP, diastolic blood pressure; DM, diabetes; eGFR, estimated glomerular filtration rate; MAP, mean arterial pressure; RCT, randomized controlled trial; and SBP, systolic blood pressure.

Across modifiers, age, baseline BP, and hypertension history accounted for the largest proportion of between-study heterogeneity. Subgroup forest plots by age, sex, BMI, and baseline SBP (Figures S2 through S6) showed patterns consistent with these findings, supporting the robustness of the identified effect modifiers.

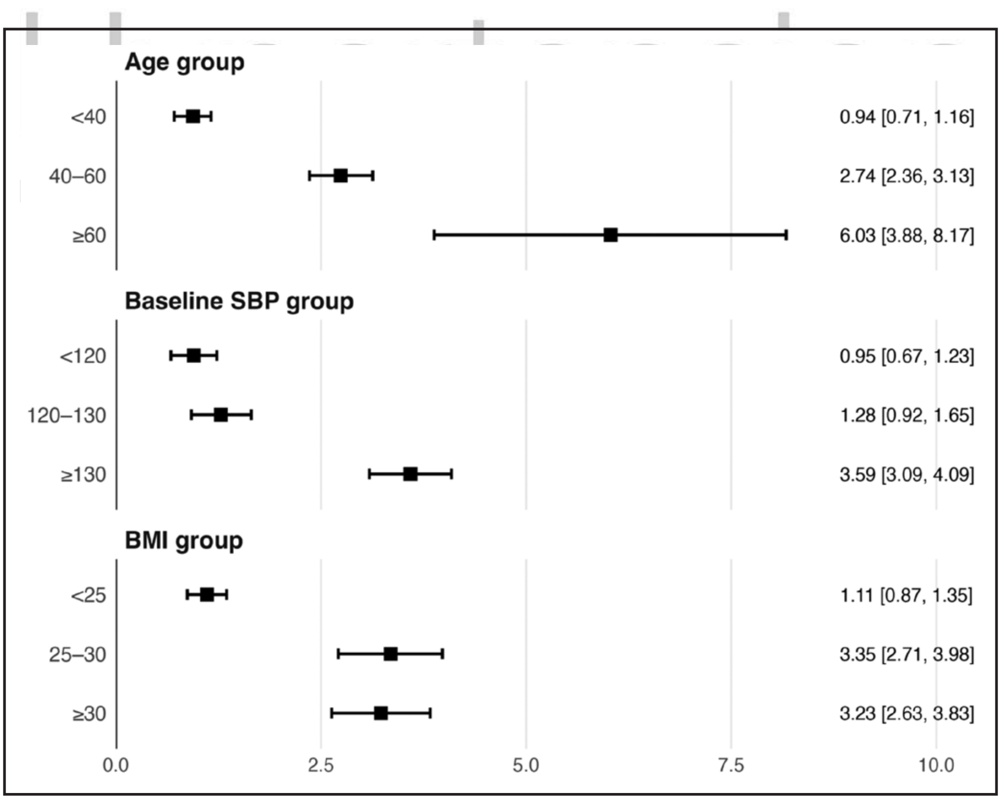

Figure 3. Stratified systolic blood pressure (SBP) response to sodium change according to age, baseline SBP, and body mass index (BMI).
Forest plot showing the estimated SBP response to sodium change across key clinical subgroups. Stratified estimates are presented according to age group, baseline SBP, and BMI. Squares represent point estimates, and horizontal lines represent 95% CIs. Larger SBP responses were observed among older individuals, those with higher baseline SBP, and individuals with higher BMI. Estimates are expressed as SBP change per 50-mmol/d change in sodium.

Simulation of the Population Distribution of Salt Sensitivity

The simulated population distribution of salt sensitivity was not symmetrical but right-skewed, indicating that while the median response was modest, a subset of individuals exhibited markedly exaggerated BP rises (Figure S8). Stratification by baseline SBP revealed a distinct distributional shift: in individuals with SBP <130 mm Hg, the distribution was narrow and centered near zero (median, 0.99 [interquartile range, −0.33 to 2.81] mm Hg). In contrast, the distribution for those with SBP ≥130 mm Hg was flattened, right-shifted, and significantly more dispersed (median, 3.48 [interquartile range, 0.79–7.11] mm Hg; Figure S9). This indicates that elevated baseline BP is associated not only with a higher mean response but also with substantially greater interindividual variability, creating a fatter tail of high-risk individuals.

When defining a highly salt-sensitive phenotype (ΔSBP ≥3 mm Hg per 50-mmol Na/d), the prevalence followed a steep, dose-dependent gradient across risk groups (Figure 4). The proportion of highly salt-sensitive individuals rose approximately >4-fold with age (from 12.5% in those aged <40 years to 56.5% in those aged >60 years) and >3-fold across baseline SBP categories (from 16.3% in those <120 mm Hg to 54.1% in those ≥130 mm Hg). Similar graded increases were observed with rising BMI (26.4% in <25 kg/m^2 versus 53.4% in ≥30 kg/m^2; all $P_{\text{for-trend}}$<0.001). These findings demonstrate that the projected prevalence of higher sodium responsiveness was greater in older, hypertensive, and higher-BMI strata.

Multivariable Metaregression

In multivariable models adjusting for age, sex, BMI, baseline SBP, and Asian proportion, age and baseline SBP remained the strongest independent modifiers of salt sensitivity (Table S2). After mutual adjustment, higher age was associated with a significantly larger SBP response to sodium change (β range, 0.66–0.71 per SD), while baseline SBP showed an even greater independent effect (β range, 0.93–1.07 per SD). BMI contributed modestly in models without SBP but was attenuated once baseline BP was included. Female proportion and Asian proportion did not significantly alter the sodium-SBP association after adjustment.

Results were highly consistent in analyses using multiply imputed covariates (Table S3). Model calibration was further assessed using observed versus predicted plots (Figure S10). Across models with and without multiple imputation, predicted and observed changes in SBP showed good agreement over the full range of values, with no evidence of systematic miscalibration. Pairwise correlations and variance inflation factor values confirmed low collinearity among included variables (all variance inflation factors <3; Table S4), supporting the stability of the multivariable models.

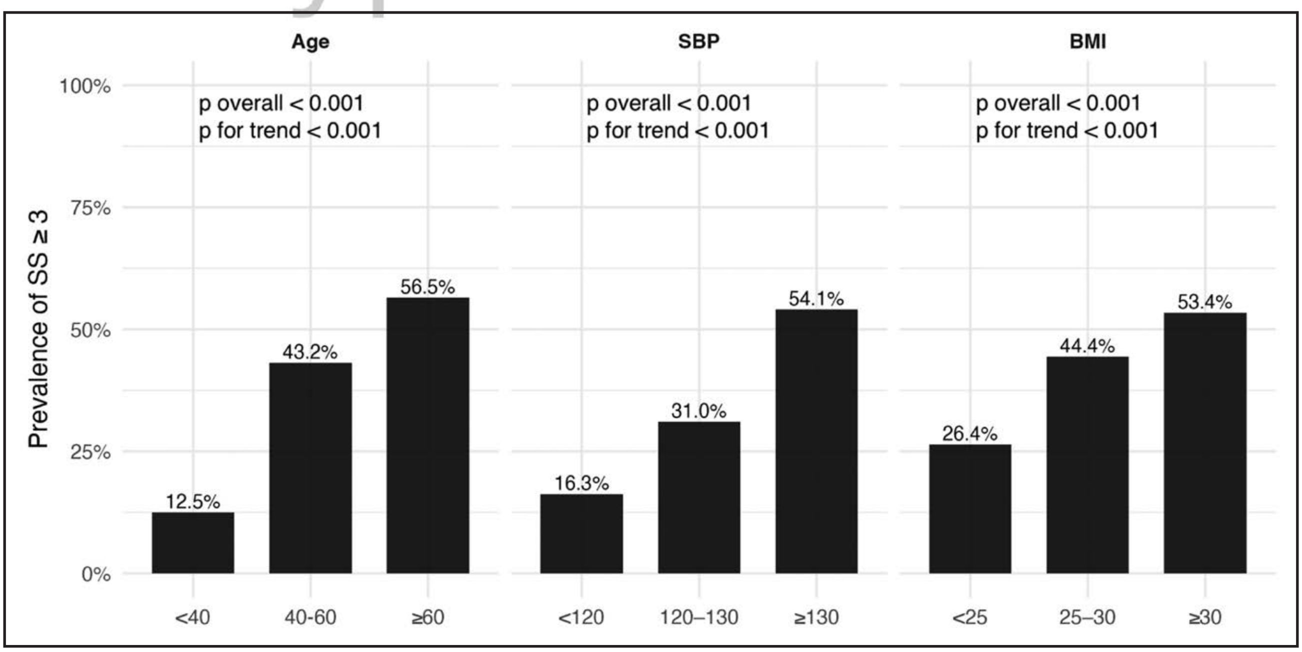

Figure 4. Prevalence of salt-sensitive phenotype (defined as salt sensitivity ≥3 mm Hg per 50-mmol Na/d) in key subgroups (simulated population).

Bar plots showing the estimated prevalence of the salt-sensitive phenotype across subgroups defined by age, systolic blood pressure (SBP), and body mass index (BMI) in a simulated population. Salt-sensitive phenotype was defined as salt sensitivity ≥3 mm Hg per 50-mmol/d change in sodium. The prevalence of the salt-sensitive phenotype increased progressively with older age, higher SBP, and higher BMI. Overall, between-group differences and trend tests were statistically significant across all 3 subgroup domains.

Model-Based Risk-Stratified Predictions Under WHO-Aligned Sodium Reduction

Predicted SBP responses were derived from the multivariable metaregression model and are presented per 50-mmol/d sodium reduction (Table 2). Sodium responsiveness varied markedly across joint strata of age, baseline SBP, and BMI. The lowest predicted response was observed in younger individuals with lower SBP and normal BMI (age <60 years, SBP <130 mm Hg, and BMI <25 kg/m^2), with a ΔSBP of 1.74 (95% CI, 1.07–2.41) mm Hg per 50 mmol/d. In contrast, the highest predicted response was observed in older individuals with elevated SBP and higher BMI (age ≥60 years, SBP ≥130 mm Hg, and BMI ≥25 kg/m^2), with a ΔSBP of 4.61 (95% CI, 3.52–5.70) mm Hg, representing more than a 2.6-fold gradient in sodium responsiveness.

For cardiovascular risk translation, ΔSBP estimates were scaled to reflect a policy-relevant sodium reduction of 100.4 mmol/d (corresponding to a reduction from current global intake to the WHO target) and converted to relative risk assuming a 2% lower cardiovascular disease risk per 1-mm Hg lower SBP. Marked heterogeneity in absolute benefit emerged. The NNR declined from 1334 in the lowest-response stratum to 87 in the highest-response stratum, indicating a >15-fold difference in projected 10-year absolute benefit. A sensitivity analysis using an SBP threshold of 140 mm Hg yielded similar results, with the NNR ranging from 2346 in the lowest-response stratum to 143 in the highest-response stratum, representing an ≈16-fold difference in projected cardiovascular risk reduction (Table S5).

A similar graded pattern was observed in the supplementary age×SBP prediction matrix (Table S6). Predicted ΔSBP ranged from 1.21 (95% CI, 0.42–1.99) mm Hg in individuals aged <45 years with SBP <120 to 6.12 (95% CI, 4.16–8.09) mm Hg in those aged ≥75 years with SBP ≥160 mm Hg, reinforcing the broad joint influence of age and baseline SBP on sodium responsiveness.

Assessment of Publication Bias

Funnel plots (Figure S11) demonstrated no meaningful small-study effects for ΔSBP or ΔDBP (Egger *P*=0.754 and 0.978), while ΔMAP showed mild asymmetry (*P*=0.040), largely driven by a few imprecise studies. Trim-and-fill correction had negligible impact on the pooled estimates.

Potassium Analyses

Among 231 effect estimates, 55 (24%) reported change in urinary potassium excretion data, enabling supplementary potassium analyses (Table S7). Concurrent urinary potassium change did not significantly modify the sodium-SBP slope (β=−0.005 mm Hg per 50-mmol/d ΔNa per 1-mmol/d ΔK [95% CI, −0.038 to 0.029]; *P*=0.787), and urinary potassium change showed no significant independent main effect (β=0.050 [95% CI, −0.133 to 0.234]; *P*=0.585).

In stepwise covariate adjustment models, the sodium-SBP association was not materially attenuated after adding urinary potassium change (step 1, ΔNa only: β=1.52 mm Hg per 50 mmol/d; step 2, ΔNa+ΔK: β=1.48; <3% attenuation). In the fully adjusted model that simultaneously includes age, BMI, baseline SBP, and urine potassium change, the ΔNa coefficient was 2.04 mm Hg (95% CI, 1.49–2.59; *P*<0.001), which is nearly identical to the primary model excluding potassium (β=1.98). The urinary potassium change term was not a significant independent predictor in any model (*P*=0.239 in the full

Table 2. Predicted SBP Response to Sodium Reduction and Corresponding Cardiovascular Risk Metrics Across 8 Age-SBP-BMI Risk Strata

Risk stratum	Age_rep, y	SBP_rep, mm Hg	BMI_rep, kg/m^2	Predicted ΔSBP per 50-mmol Na/d, mm Hg*	RR	10-year CVD risk (by SCORE2)	NNR
Age <65 y | SBP <130 | BMI <25	44	117.8	23.0	1.74 (1.07–2.41)	0.932	1.10%	1334
Age <65 y | SBP <130 | BMI ≥25	44	117.8	27.4	2.15 (1.29–3.00)	0.916	1.10%	1089
Age <65 y | SBP ≥130 | BMI <25	44	142.5	23.0	2.55 (1.60–3.50)	0.902	1.50%	679
Age <65 y | SBP ≥130 | BMI ≥25	44	142.5	27.4	2.96 (2.19–3.72)	0.887	1.50%	589
Age ≥65 y | SBP <130 | BMI <25	65	117.8	23.0	3.39 (1.69–5.09)	0.871	5.30%	147
Age ≥65 y | SBP <130 | BMI ≥25	65	117.8	27.4	3.80 (2.27–5.33)	0.857	5.30%	132
Age ≥65 y | SBP ≥130 | BMI <25	65	142.5	23.0	4.20 (2.67–5.73)	0.843	6.80%	94
Age ≥65 y | SBP ≥130 | BMI ≥25	65	142.5	27.4	4.61 (3.52–5.70)	0.829	6.80%	87

Green shading, low-risk strata (NNR >1200); yellow shading, moderate-risk strata (NNR=200–1200); orange shading, high-risk strata (NNR <200). Predicted ΔSBP per 50-mmol/d sodium reduction was derived from the multivariable metaregression model (Table S2, model 5; k=231 effect estimates). ΔSBP values were linearly rescaled to 100 mmol/d for the NNR calculation, reflecting a policy-relevant reduction from estimated global mean sodium intake (≈187 mmol/d) to the WHO 87-mmol/d target. Ten-year CVD risk was estimated using the SCORE2 model (calibrated to a moderate-risk European population) using stratum-representative age and SBP values. NNR=1/(baseline 10-year CVD risk×[1−RR]). RR after sodium reduction=0.98^ΔSBP_100. BMI indicates body mass index; CVD, cardiovascular disease; NNR, number needed to restrict; RR, predicted relative risk after sodium restriction; and SBP, systolic blood pressure.

*ΔSBP values were rescaled to 100 mmol/d for NNR calculation (see Methods section).

model). Among effect estimates with the urinary sodium-to-potassium ratio, it was not significantly associated with SBP change (β=−0.118 [95% CI, −0.310 to 0.074]; *P*=0.216), likely reflecting limited power and heterogeneous intervention protocols within this subset. These findings indicate that concurrent potassium changes do not materially confound the sodium-BP associations in this analysis.

We additionally examined whether baseline urinary potassium excretion, as a proxy for habitual dietary potassium intake at the study level, modified salt sensitivity. Of 114 effect estimates with accessible baseline urinary potassium data (mean, 57.2±20.2 mmol/d), elevated baseline urinary potassium exhibited a marginal trend towards diminished sodium-SBP responsiveness (β=−0.171 mm Hg per 50-mmol/d ΔNa per 10-mmol/d increase in baseline urinary K [95% CI, −0.362 to 0.019]; *P*=0.077; R2 analog=9.5%) although this did not achieve statistical significance, presumably due to restricted statistical power. The direction of this association is consistent with the biological expectation that higher habitual potassium intake attenuates sodium-induced BP elevation.

Post-JNC7 Era Sensitivity Analysis

In a sensitivity analysis restricted to post-JNC7 studies (year ≥2003; k=131 effect estimates), the pooled sodium-SBP slope was β=1.783 mm Hg per 50 mmol/d (95% CI, 1.532–2.034; I^2=75.0%), identical to the primary analysis (β=1.783 [95% CI, 1.568–1.998]; I^2=76.2%). Results were consistent with a 2005 publication threshold (β=1.787 [95% CI, 1.522–2.053]). The 3 key effect modifiers, age, baseline SBP, and BMI, remained highly significant and directionally consistent in all temporal subsets (all statistically significant; Table S8).

DISCUSSION

In this most comprehensive and up-to-date systematic review with meta-analysis and metaregression analysis, encompassing 160 studies and 16 443 participants, each 50-mmol/d increment in urinary Na excretion was independently associated with a 1.76-mm Hg higher SBP. Salt sensitivity varied systematically across populations: advancing age and higher baseline SBP emerged as the dominant effect modifiers, with additional, albeit more modest, contributions from elevated BMI and female proportion. When expressed as a clinically interpretable binary phenotype, marked salt sensitivity prevalence was higher in older adults with established hypertension or obesity. These findings suggest that readily measured clinical characteristics may help identify populations with greater expected BP responsiveness to sodium reduction and further guide sodium-restricting policy in populations with different risks.

The substantial between-study heterogeneity observed in ΔSBP (I^2=76.2%) should be interpreted as a biological signal rather than a statistical limitation, reflecting genuine variation in sodium-BP coupling across populations. That age and baseline SBP accounted for a substantial proportion of this heterogeneity strengthens their role as pragmatic stratification criteria. The residual unexplained heterogeneity, however, likely reflects individual-level variation that population-level aggregate data cannot fully resolve, underscoring the need for future IPD meta-analyses to further disentangle the determinants of salt sensitivity at the person level.

Our findings have potential translational relevance for clinical counseling and public health policy prioritization. At the population level, sodium-BP responsiveness appeared greater in studies enrolling older participants and those with higher baseline SBP or BMI, and they are, thus, more likely to translate into clinically meaningful BP changes. Identifying these characteristics does not require specialized testing, yet they may serve as pragmatic indicators of heightened susceptibility.

More than 15-fold gradient in the NNR across age, baseline SBP, and BMI strata, from ≈1334 in younger, normotensive, lean individuals to ≈87 in older, hypertensive, overweight adults (Table 2), has direct implications for how sodium restriction guidance should be communicated and prioritized. In older adults with hypertension and overweight or obesity, an NNR of ≈87 over 10 years is comparable to established pharmacological therapies for secondary cardiovascular prevention, positioning sodium restriction as one of the most efficient nonpharmacological interventions available in this population. By contrast, the NNR exceeding 1200 in younger, normotensive, lean individuals does not preclude population-wide sodium-reduction messaging, whose public health rationale rests on population-attributable risk rather than individual-level NNR, but does suggest that intensive, individualized counseling resources are most efficiently directed toward higher-risk strata. While both the 2024 European Society of Cardiology and the 2025 American Heart Association/American College of Cardiology guidelines advocate a uniform dietary sodium target (<87–100 mmol/d) without stratification by individual susceptibility, our data challenge the adequacy of this undifferentiated approach. Using 3 routinely available clinical parameters, age, baseline SBP, and BMI, we identified a 16-fold gradient in projected absolute 10-year cardiovascular benefit across risk strata, a magnitude of heterogeneity not currently acknowledged in guideline recommendations.

A principal barrier to the clinical adoption of individualized sodium targets has been the absence of a practical, scalable method for identifying highly salt-sensitive individuals without labor-intensive sodium-loading protocols. The present analysis suggests that this barrier is lower than previously assumed. Age, baseline SBP, and BMI, universally recorded clinical parameters, identify strata that differ strikingly in both the prevalence of marked salt sensitivity and in projected absolute cardiovascular benefit (NNR ≈1334–≈87; >15-fold). Because these variables are already embedded in routine care, personalized sodium counseling requires no additional testing, only systematic use of existing clinical data. This represents a pragmatic screening framework analogous to traditional cardiovascular risk stratification.

Prior meta-analyses have quantified the mean sodium-BP dose-response relationship but have not translated this relationship into the population-level prevalence of clinically meaningful responders across risk strata. In the most methodologically rigorous predecessor analysis by Huang et al,[24] which was restricted to randomized trials with urinary sodium verification, each 50-mmol/d reduction in sodium excretion was associated with a 1.10-mm Hg fall in SBP[24]; intervention duration substantially modified the estimate, with trials longer than 15 days yielding approximately double the reduction of shorter trials. Our pooled estimate (1.78 mm Hg per 50 mmol/d; equivalent to 3.56 mm Hg per 100 mmol/d) exceeds the overall estimate reported by Huang et al (2.2 mm Hg per 100 mmol/d) but falls within the range of their duration-stratified subgroup analyses (1.4–5.1 mm Hg per 100 mmol/d). This discrepancy is methodologically informative: by spanning a broader spectrum of study designs, exposure durations, and populations, the present analysis captures the full distributional range of sodium-BP coupling encountered in clinical practice, rather than the more conservative estimate obtained from trials with urinary sodium verification alone. Compared with prior meta-analyses restricted to randomized trials with urinary sodium verification, our broader inclusion criteria allowed exploration of heterogeneity across a wider range of populations and study designs. This broader scope may partly explain the larger pooled estimate and greater heterogeneity observed here.

The strong modification of salt sensitivity by age and baseline BP observed in this analysis is biologically plausible and consistent with established models of BP regulation. Aging is accompanied by progressive vascular stiffening, endothelial dysfunction, and reduced arterial compliance, which increase the dependence of SBP on intravascular volume and sodium balance.[25–27] With advancing age, renal sodium handling becomes progressively impaired, characterized by reductions in renal blood flow and functional nephron mass, together with a rightward shift of the pressure-natriuresis relationship.[28,29] These changes limit the kidney's ability to excrete sodium without concomitant increases in BP, rendering systemic BP more sensitive to variations in sodium intake. Declining levels of antiaging factors, such as klotho, may further exacerbate these processes by promoting salt-induced hypertension and impairing vascular and renal resilience.[30,31] Baseline BP likely integrates these cumulative renal and vascular alterations, explaining its strong association with steeper sodium-BP slopes and higher prevalence of salt-sensitive phenotypes.[32] Together, these mechanisms position aging as a central determinant of salt sensitivity.

Excess adiposity is characterized by expansion of extracellular fluid volume, enhanced renal sodium reabsorption, and heightened sympathetic nervous system activity.[33,34] Adipose-derived factors, including leptin and aldosterone-related mediators, have been implicated in promoting epithelial sodium channel activity and sodium retention, while obesity-related low-grade inflammation and metabolic dysfunction may further impair renal sodium handling and vascular responsiveness.[35–37] These mechanisms provide biological plausibility for the higher prevalence of salt-sensitive phenotypes observed among overweight and obese populations. The attenuation of BMI effects after adjustment for baseline SBP likely reflects SBP as an integrative downstream marker of cumulative hemodynamic and metabolic burden. Together, these findings support adiposity as an important contextual modifier of salt sensitivity.

Sex differences in salt sensitivity have been variably reported. Our study-level metaregression suggests a modest association with a higher proportion of women, but residual confounding (age distribution, baseline BP, menopausal status, and study design) may contribute. Prior experimental evidence suggests that sodium-related BP responses in women may be driven less by classic pressure-natriuresis mechanisms and more by vascular pathways, including endothelial dysfunction and heightened mineralocorticoid receptor activity.[38–40] Aldosterone-mediated activation of endothelial epithelial sodium channels has been shown to impair nitric oxide bioavailability and reduce vasodilatory capacity, thereby amplifying BP responses to sodium intake.[41] In addition, sex hormones modulate renal and vascular sodium handling, with estrogen and progesterone influencing arterial stiffness, endothelial function, and adrenal responsiveness to angiotensin II.[42,43] While these mechanisms provide biological plausibility for sex-specific patterns of salt sensitivity, the study-level nature of our analyses precludes definitive separation of hormonal, vascular, and BP–related effects, highlighting the need for individual-level investigations.

We did not observe robust independent modification of salt sensitivity by ethnicity, diabetes prevalence, or baseline kidney function. Although prior individual-level studies have suggested greater salt sensitivity among Black populations and among individuals with diabetes or impaired kidney function,[44–47] the absence of consistent effects in our study-level analyses likely reflects limited statistical power due to incomplete reporting of these characteristics. The variability in treatment backgrounds among the included studies, which spans from untreated normotensives to patients on multiple antihypertensive medications, constitutes an additional source of interstudy variability that IPD analyses could more effectively delineate. The potential contribution of dietary potassium to the observed sodium-BP associations merits its consideration. Accumulating evidence indicates that potassium independently lowers BP,[48] and the urinary sodium-to-potassium ratio may better predict cardiovascular risk than either ion alone.[49] In our supplementary analyses, however, concurrent urinary potassium changes neither significantly modified the sodium-BP slope nor materially attenuated the sodium-BP coefficient after adjustment. These findings suggest that, within the available literature, the sodium-BP gradient reported here is not primarily an artifact of co-occurring potassium changes. Nonetheless, changes in urinary potassium excretion data were available in only 55 of 231 effect estimates (24%), which limits the power of these analyses. A borderline trend toward attenuated salt sensitivity with higher baseline urinary potassium (β=−0.171 per 10 mmol/d; *P*=0.077; k=114) further suggests that habitual potassium intake may buffer sodium-induced BP elevation though statistical power was insufficient to confirm this in the present data set. Routine measurement of urinary potassium in future sodium intervention trials would enable a more complete characterization of the independent and joint contributions of these 2 dietary ions to BP regulation. Albuminuria data were insufficient across included studies to permit metaregression analysis, despite its biological plausibility as a modifier of renal sodium handling and salt sensitivity; this should be addressed in future IPD analyses.

Strengths of this study include the large evidence base spanning diverse populations and sodium exposure ranges, standardization of effect estimates to a common sodium scale, and systematic assessment of effect modification using prespecified metaregression. We prioritized 24-hour urinary sodium when available and evaluated robustness across alternative exposure definitions and modeling assumptions. By translating continuous salt sensitivity into phenotype prevalence, we provide an interpretable framework for identifying groups in whom sodium reduction may yield larger BP benefits. In addition, the use of metaregression and extensive sensitivity analyses, including Knapp-Hartung adjustment and multiple imputation, supports the robustness of our findings.

This study has limitations. First, sodium exposure assessment and intervention protocols varied across studies; although we standardized effects to a common sodium scale, measurement error may attenuate associations. Second, many interventions were short to medium in duration; long-term BP effects of sustained dietary change may differ from short-term physiological responses. Third, metaregression is based on study-level covariates and is susceptible to ecological bias; modifiers identified here should be confirmed using IPD. Fourth, phenotype prevalence was derived from model-based simulations rather than direct clinical testing, and external validation in contemporary cohorts will be important. Fifth, changes in urinary potassium excretion data were available in only 24% of the included effect estimates, restricting the power of potassium-focused analyses.

Perspectives

Dietary sodium reduction remains a cornerstone of hypertension prevention and control, but its expected benefit is unlikely to be uniform across individuals. By integrating evidence from sodium intervention and observational studies, our findings suggest that routinely available clinical characteristics, particularly age, baseline SBP, and BMI, may help identify populations with greater sodium-BP responsiveness and larger projected absolute cardiovascular benefit. These results should not be interpreted as reducing the importance of population-wide sodium reduction, which remains essential for public health, but rather as supporting a complementary precision-prevention approach in which intensive counseling and implementation efforts are prioritized for those most likely to benefit. Future IPD meta-analyses and prospective validation studies are needed to confirm these modifiers, refine prediction models of salt sensitivity, and determine whether risk-stratified sodium counseling improves adherence, BP control, and cardiovascular outcomes beyond uniform recommendations.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Text S1–S4
Tables S1–S8
Figures S1–S13

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